

Information and Communication Technology

Everyday life for employees, students, consumers, savers, investors, managers, and entrepreneurs is changing and becoming more complex all the time. Although rapid changes in information and communication technology have an impact on all aspects of life, the operations of business are particularly affected. The information and communication technology courses aim to prepare students for a business and communication world that relies more and more on electronic technology, technology that changes at a rapid pace. Whether students wish to apply the new technologies in their careers in the workplace, to develop a broad understanding of the effects of technology on business, or to establish a foundation for the use of information technology at the postsecondary level, the information and communication technology courses will provide opportunities to develop relevant skills and knowledge.

Strands

Information and Communication Technology: The Digital Environment, Grade 11, Open

- Digital Literacy
- Productivity Software
- Business Communications
- E-Business
- Information and Communication Technology Ethics and Issues

Information and Communication Technology: Multimedia Solutions, Grade 12, College Preparation

- The Electronic Business Environment
- Applications Software
- Electronic Communications
- Project Management

Information and Communication Technology in the Workplace, Grade 12, Workplace Preparation

- The Electronic Workplace Environment
- Applications Software
- Electronic Communications and Teamwork
- Preparation for Employment

Information and Communication Technology: (BTX4C) Multimedia Solutions, Grade 12, College Preparation

This course provides students with the opportunity to apply their information and communication technology skills while working in a team environment. Through a project-based approach, students will have opportunities to integrate common business software applications and apply multimedia techniques. Students will further develop their understanding of electronic business and e-commerce environments. The skills acquired in this course will prepare students for success in postsecondary studies and in their future careers.

Prerequisite: Information and Communication Technology: The Digital Environment, Grade 11, Open

The Electronic Business Environment

Overall Expectations

By the end of this course, students will:

- assess workplace settings from an information and communication technology perspective;
- analyse the impact of information and communication technology on the business environment;
- assess the importance of adhering to ethical standards when conducting business electronically.

Specific Expectations

The Digital Workplace

By the end of this course, students will:

- compare a variety of workplace settings (e.g., home office, small business, large corporation) in terms of their information and communication technology needs;
- describe the software (e.g., Internet conferencing software, personal information management software) and hardware (e.g., mobile computing devices) used in a variety of workplace settings;
- analyse the ways in which workplace settings have changed as a result of recent developments in information and communication technology;
- explain terminology related to computer networks (e.g., configuration, topology, network);
- describe a variety of network topologies (e.g., star, ring, bus, tree);
- analyse appropriate equipment arrangements and layouts for specific business situations.

The Impact of Information and Communication Technology

By the end of this course, students will:

- identify different types of businesses (e.g., e-commerce, data warehouse, web host) that have evolved as a result of information and communication technology developments;

- describe the positive and negative effects of information and communication technology on businesses and working conditions (e.g., more efficient communication, ease of information sharing/retrieval, job loss/reassignment, technical difficulties);
- identify the technological challenges of doing business globally (e.g., in connection with payment, language differences, cultural differences, time zones).

Business Ethics

By the end of this course, students will:

- analyse the impact of information and communication technology on a business code of ethics;
- explain the importance of observing ethical standards when conducting business electronically;
- assess the consequences of unethical behaviour related to information and communication technology for an employee and for a business.

Applications Software

Overall Expectations

By the end of this course, students will:

- use appropriate software to create integrated customized documents that meet professional business standards;
- demonstrate effective use of multimedia software;
- use appropriate software to perform a variety of business tasks.

Specific Expectations

Integrating Applications

By the end of this course, students will:

- identify the most appropriate software applications for specific tasks;
- demonstrate appropriate use of the advanced software features and functions required for creating business documents (e.g., styles, generated table of contents, embedded and linked objects, macros, pivot tables, forms);
- produce integrated business documents (e.g., reports with charts, mail merge letters using data in an existing data source, linked documents in a presentation) for specific purposes.

Multimedia

By the end of this course, students will:

- differentiate between multimedia file formats (e.g., GIF, JPG, PNG, WAV, AVI) and use a variety of these formats when developing multimedia products;

- use graphic editing software to edit (e.g., crop; adjust colour, size) and export graphics;
- use advanced multimedia features (e.g., sound, animation, video, graphics) to enhance a product (e.g., web page, presentation, e-newsletter).

Business Productivity Software

By the end of this course, students will:

- demonstrate the effective use of a variety of applications software (e.g., database, financial planning tools, graphic editing, animation, audio editing);
- complete simulated or real tasks using appropriate applications software features;
- use applications software reference materials (e.g., online help, vendor websites, tutorials, manuals) to enhance their software skills.

Electronic Communications

Overall Expectations

By the end of this course, students will:

- use a variety of electronic resources to retrieve, evaluate, and communicate information;
- demonstrate an understanding of the importance of keeping an updated digital portfolio that accurately reflects their information and communications skills and competencies;
- demonstrate an understanding of e-commerce solutions.

Specific Expectations

Electronic Research and Communication

By the end of this course, students will:

- explain how search engines work and how they rank results;
- evaluate information collected from electronic sources for usefulness, validity, bias, accuracy, reliability, and currency;
- adhere to copyright and licensing rules and regulations when accessing information electronically;
- synthesize information from a variety of sources (e.g., websites, online magazine articles, CD-ROMs, databases);
- select and use appropriate electronic tools (e.g., presentation software, e-mail, web-based communication) to communicate information to a specific audience;
- communicate research results electronically.

Digital Portfolio

By the end of this course, students will:

- identify employability skills and essential skills by accessing appropriate websites (e.g., Conference Board of Canada, Human Resources and Social Development Canada), and assess and summarize these skills;
- classify their information and communication technology skills (e.g., skills in web-page design, creating multimedia presentations, desktop publishing) and include a summary of them in their portfolio;

- demonstrate their information and communication technology skills in samples of their work;
- select relevant information and appropriate samples of their work to be included in a digital portfolio that can be used for a variety of purposes (e.g., job search, school admission, program admission, job promotion).

E-Commerce Solutions

By the end of this course, students will:

- identify features of payment gateways for e-commerce (e.g., merchant account, real-time credit authorization, off-line order processing);
- identify methods used by consumers to pay for online purchases (e.g., smart card, digital wallet, person-to-person payment);
- summarize, on the basis of research, different methods of advertising used on websites to generate revenue (e.g., banner ads, interstitial ads, sponsorship/partnership ads);
- describe security measures used on websites (e.g., SSL, encryption, digital certificates, digital signatures);
- analyse privacy policies on e-commerce websites.

Project Management

Overall Expectations

By the end of this course, students will:

- demonstrate an understanding of a team-based approach to project management;
- demonstrate the use of electronic tools to manage a multimedia team project;
- use a team-based approach to create a multimedia product.

Specific Expectations

Team-Based Project Management

By the end of this course, students will:

- identify advantages of a team-based approach to project management (e.g., collaboration, delegation of roles based on individual expertise);
- compare effective and ineffective teams and explain how they differ;
- identify challenges and advantages of working in a culturally and socially diverse workplace (e.g., bridging differences in customs and religion and ethnocultural backgrounds; access to a range of cultural and educational experiences);
- describe how businesses resolve employee conflict to enhance productivity.

Electronic Project Management Tools

By the end of this course, students will:

- identify and describe the electronic tools used by business to facilitate project-team activities (e.g., e-mail, intranet, software features that enhance collaboration, scheduling tools, fax, video conferencing);
- describe applications software that is specifically designed to facilitate project management;
- use appropriate electronic tools to enhance team productivity.

Multimedia Product

By the end of this course, students will:

- implement an appropriate project-management process to complete a team project (e.g., set goals, organize, schedule, budget, control, evaluate);
- create a multimedia product (e.g., school website, e-commerce website, presentation, animation) in a team environment, using appropriate software;
- use effective strategies (e.g., conflict resolution, decision making) to enhance team productivity;
- communicate the finished product to an appropriate audience, using effective presentation techniques.